

Requirement Specification

Mobile Device E-Commerce OMS

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1. Introduction

1.1 Purpose

This Business Requirement Specification (BRS) defines the functional and non-functional requirements for an E-Commerce Order Management System (OMS) specifically tailored for a mobile device retail

business. Due to the high value and targeted nature of smartphones and electronics, this system prioritizes strict serial number tracking, security, and fraud prevention.

1.2 Scope

The system will manage the lifecycle of mobile device orders, encompassing inventory sync, IMEI/Serial Number tracking, bundle management (e.g., phone + accessories), secure shipping logic, and rigorous reverse logistics for electronic returns.

2. Functional Requirements

2.1 High-Value Inventory Management

Electronics require unique identification capabilities beyond standard SKU counts.

Req ID	Description	Priority
FR-INV-01	The system shall support IMEI and Serial Number tracking for every individual mobile device unit received, stored, and dispatched.	High
FR-INV-02	The system shall automatically register the dispatched IMEI with the corresponding manufacturer's warranty database upon order fulfillment.	Medium
FR-INV-03	The system shall support dynamic bundling (e.g., deducting 1 smartphone, 1 case, and 1 screen protector when a promotional kit is purchased).	High

2.2 Order Processing & Fraud Prevention

Due to the high resale value of mobile phones, fraud detection is a critical path in order processing.

Req ID	Description	Priority
FR-ORD-01	The system shall integrate with a 3rd-party fraud detection API to flag high-risk transactions (e.g., mismatched billing/shipping, unusual IP velocity).	High
FR-ORD-02	Orders flagged for medium or high fraud risk must automatically route to a manual review queue, pausing inventory allocation.	High
FR-ORD-03	The system shall enforce purchase limits on specific high-demand SKUs per customer account or shipping address.	Medium

2.3 Secure Fulfillment & Dispatch

Shipping logistics must account for secure transit and delivery verification.

Req ID	Description	Priority
FR-SHP-01	The system shall mandate "Signature Required" and package insurance options on all shipping labels generated for orders exceeding \$500.	High
FR-SHP-02	The packing workflow must enforce an mandatory barcode scan of the device's IMEI before generating the final shipping label.	High

2.4 Electronics Reverse Logistics (Returns)

Handling returns for electronics involves verifying device condition and activation locks.

Req ID	Description	Priority
FR-RET-01	The system must prompt the customer to acknowledge the removal of activation locks (e.g., iCloud, Google Account) before generating an RMA label.	High
FR-RET-02	Warehouse receiving staff must input a multi-point inspection result (screen condition, water damage, IMEI match) before releasing funds for refund.	High
FR-RET-03	The system shall automatically calculate and apply restocking fees if the inspection indicates the device was opened or used.	Medium

3. Non-Functional Requirements

3.1 Security & Compliance

- **Data Protection:** All customer data and associated IMEI histories must be encrypted to prevent data breaches that could compromise device security.
- **Access Control:** Warehouse staff shall only have access to view necessary order details; full financial data must be masked at the packing station.

3.2 Integration & Performance

- **APIs:** Must feature seamless webhook integrations for major Telecom carriers (for SIM activation tracking) and localized warranty databases.
- **Reliability:** The platform must maintain high availability (99.99%) to ensure no disruptions during major device launch days (e.g., new iPhone releases).